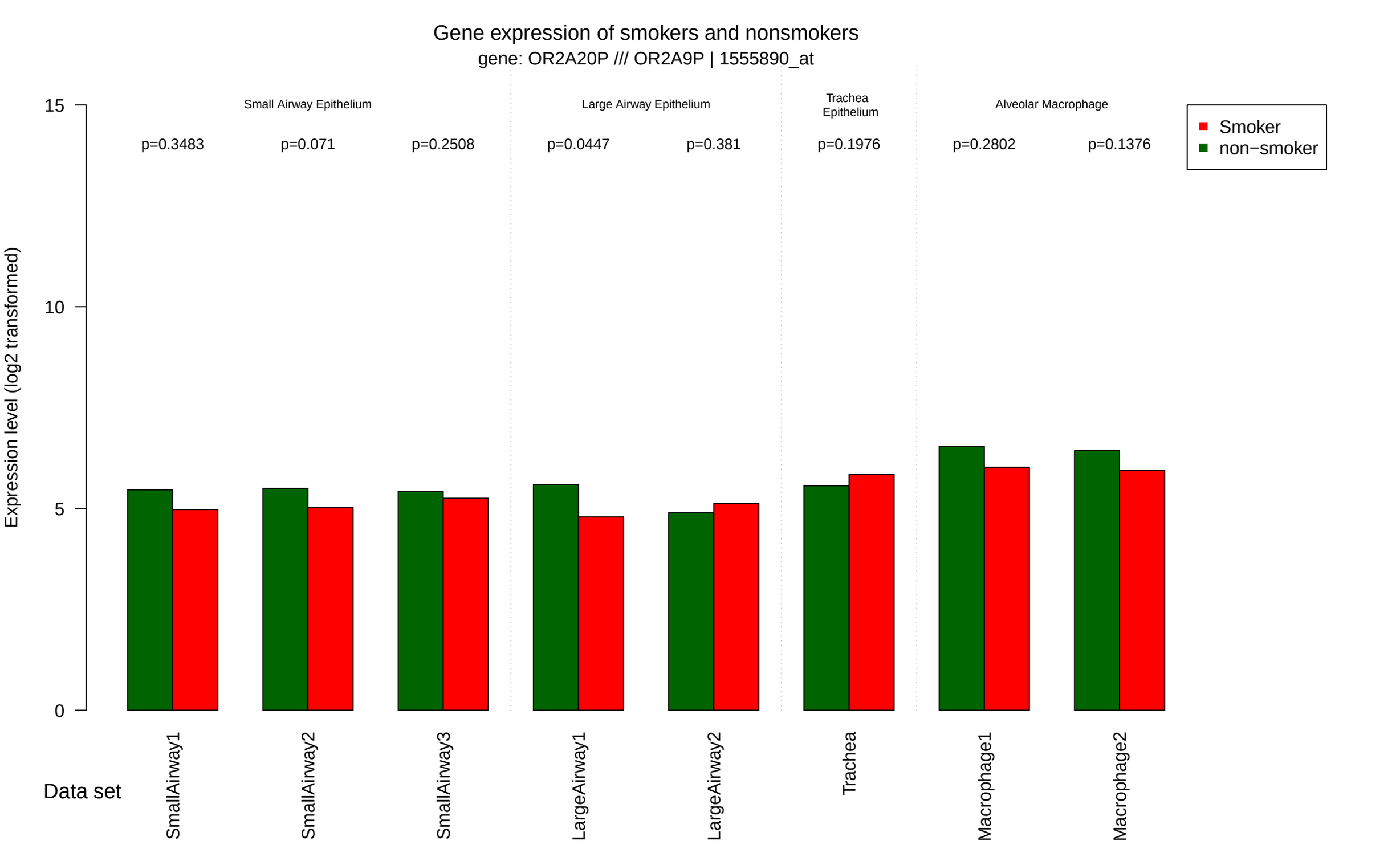
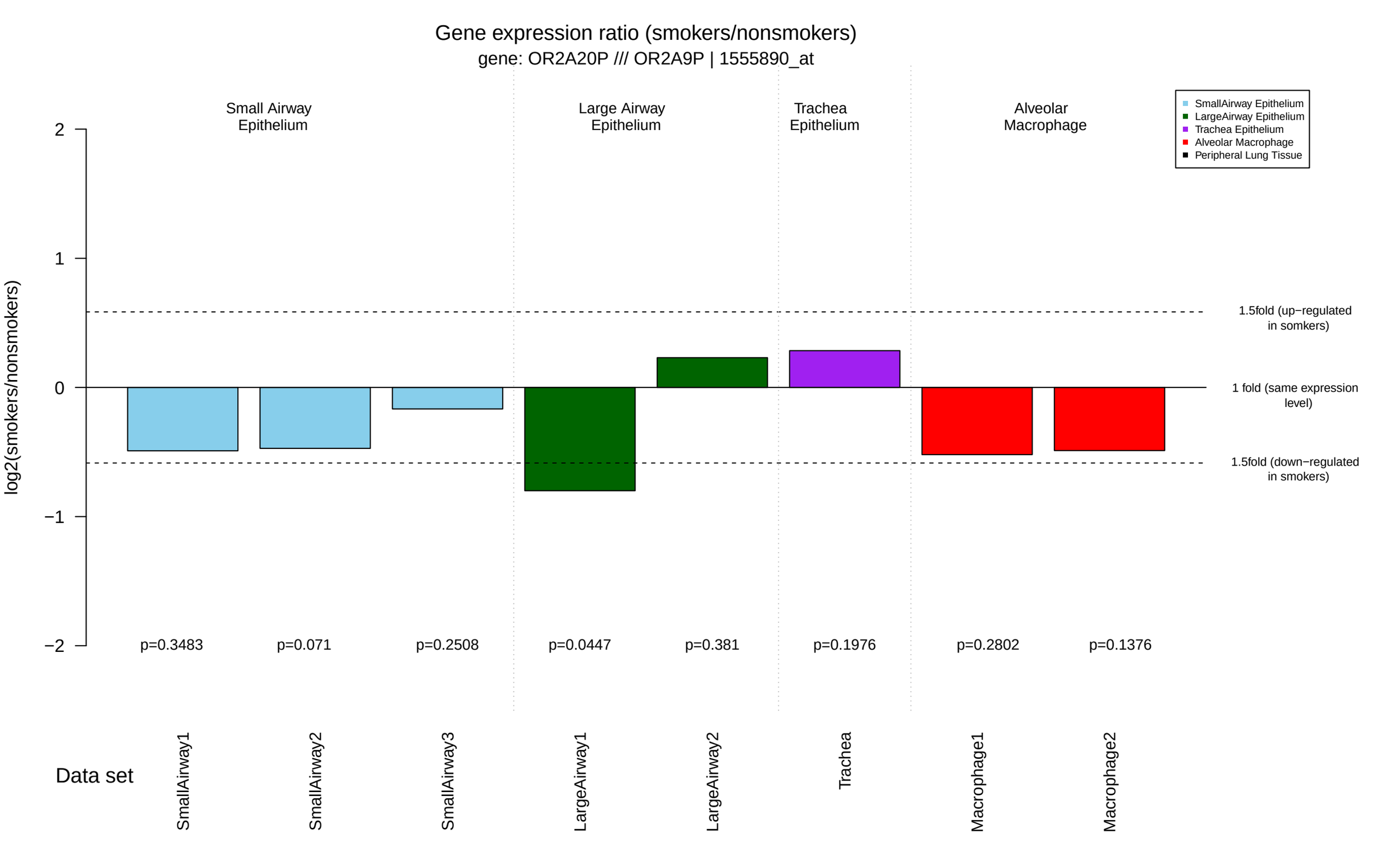
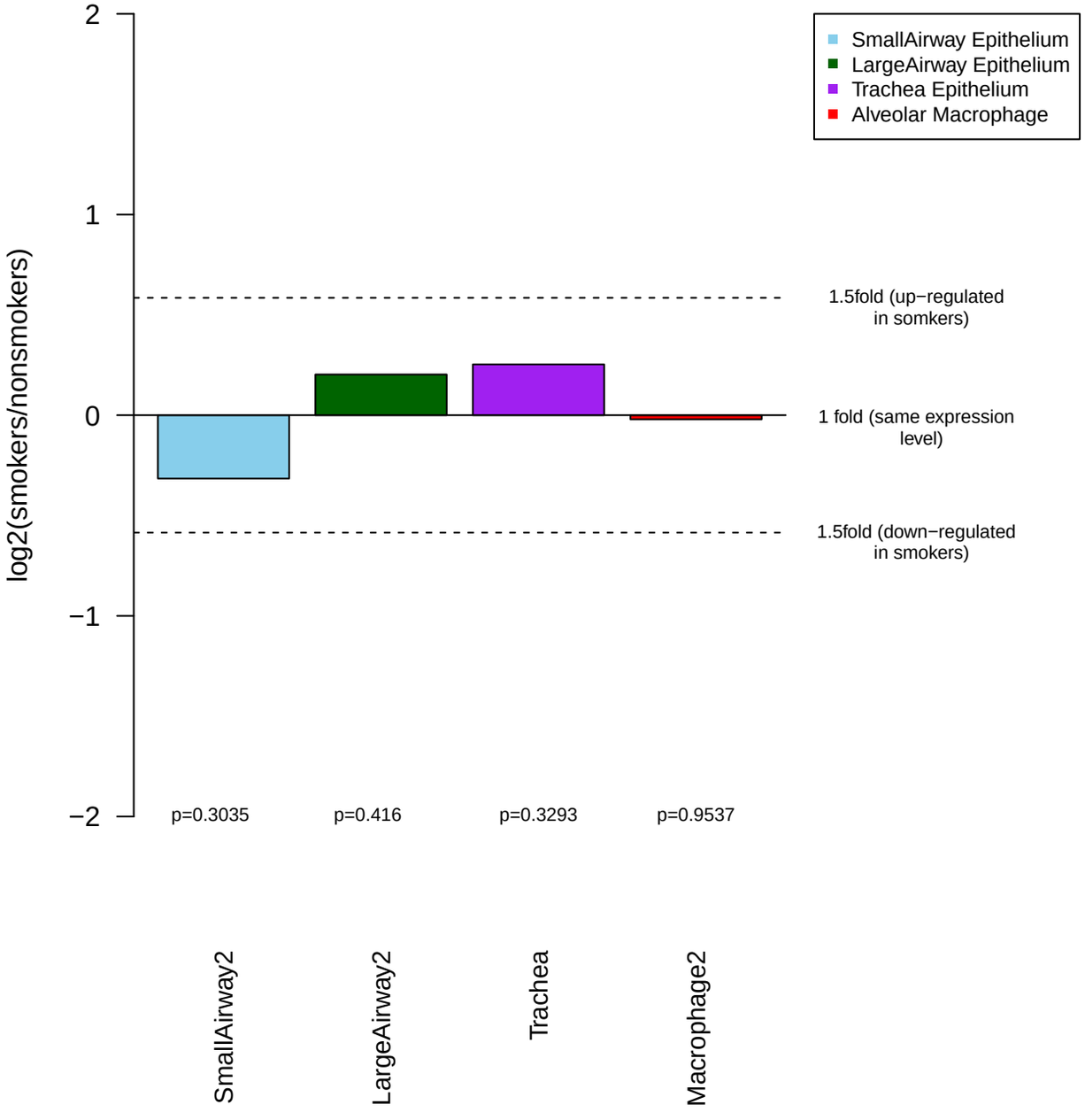

Gene expression of smokers and nonsmokers
gene: OR2A20P /// OR2A9P | 1555890_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3483
p=0.071
p=0.2508
p=0.0447
p=0.381
p=0.1976
p=0.2802
p=0.1376
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: OR2A20P /// OR2A9P | 1555890_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3483
p=0.071
p=0.2508
p=0.0447
p=0.381
p=0.1976
p=0.2802
p=0.1376
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

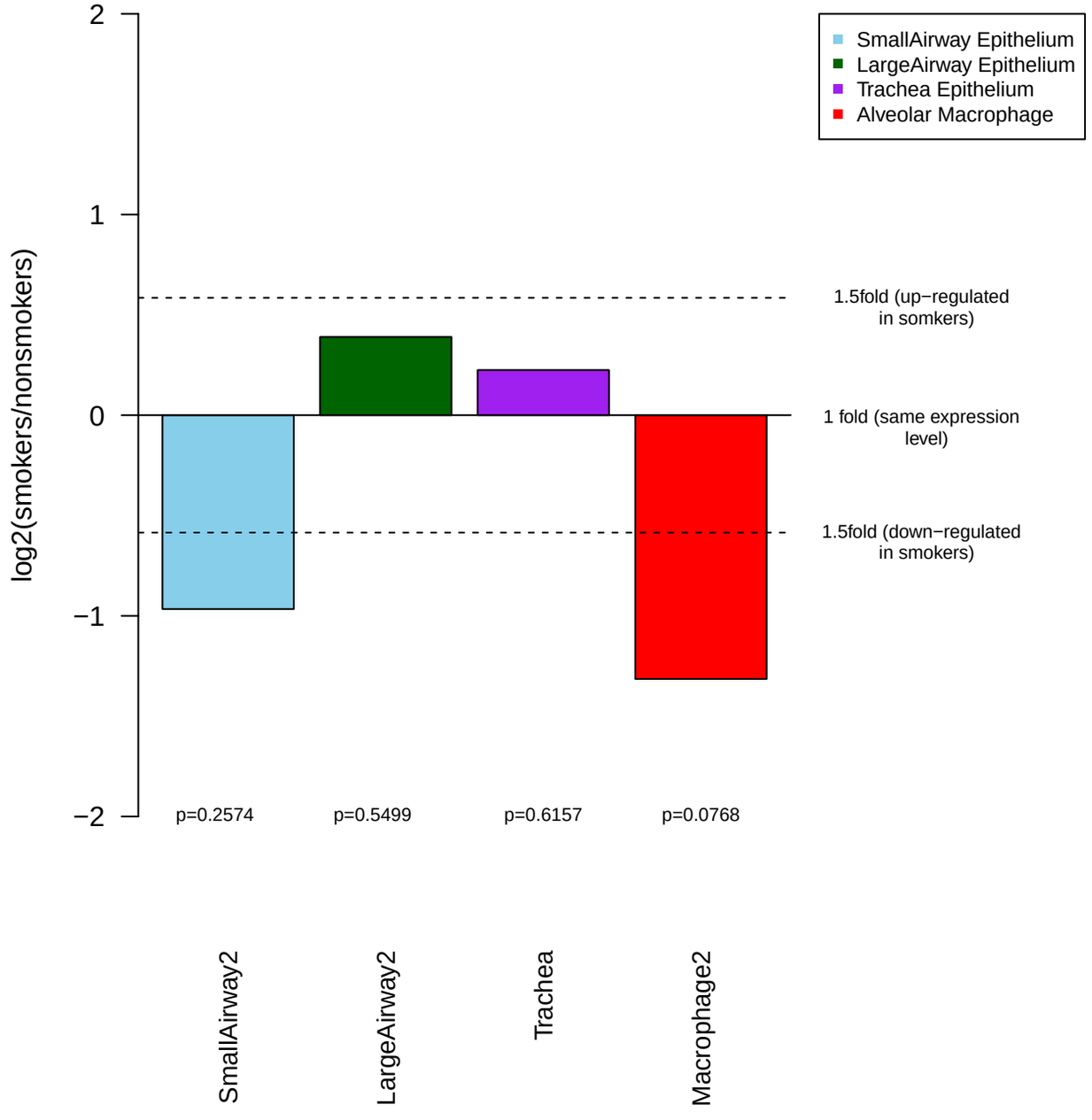
Gene expression ratio in Males

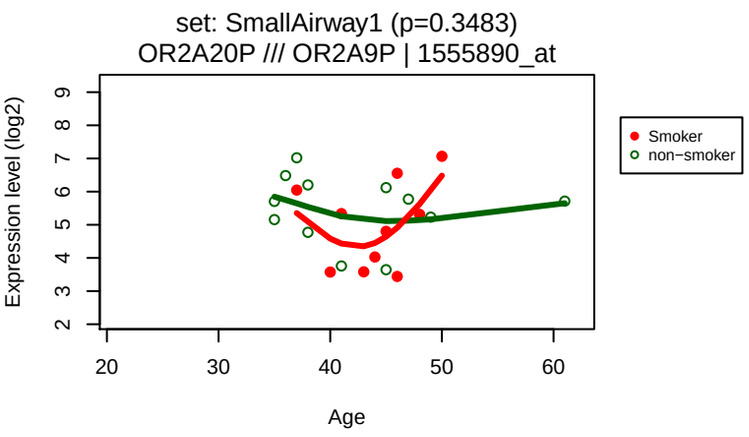
gene: OR2A20P /// OR2A9P | 1555890_at

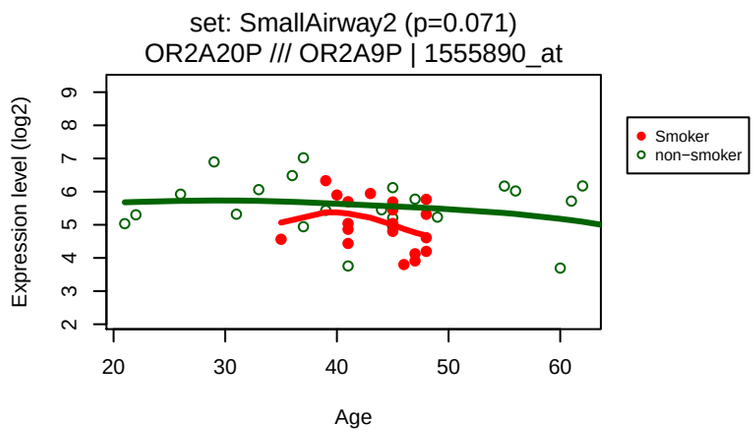


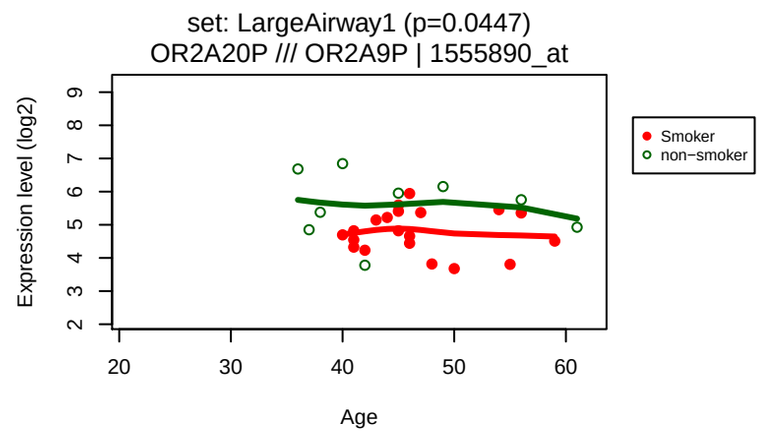
Gene expression ratio in Females

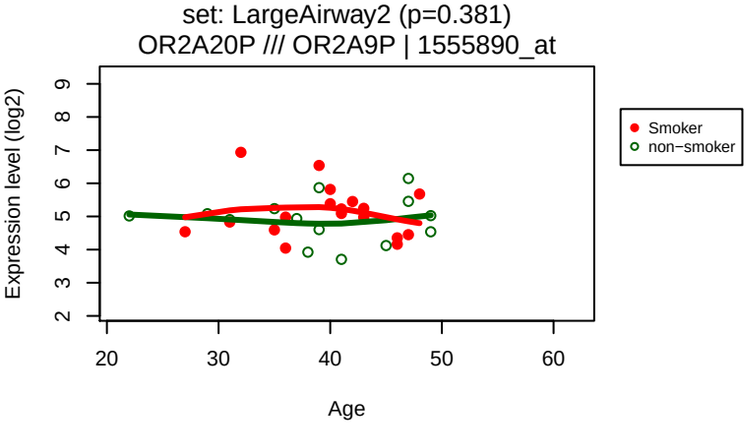
gene: OR2A20P /// OR2A9P | 1555890_at

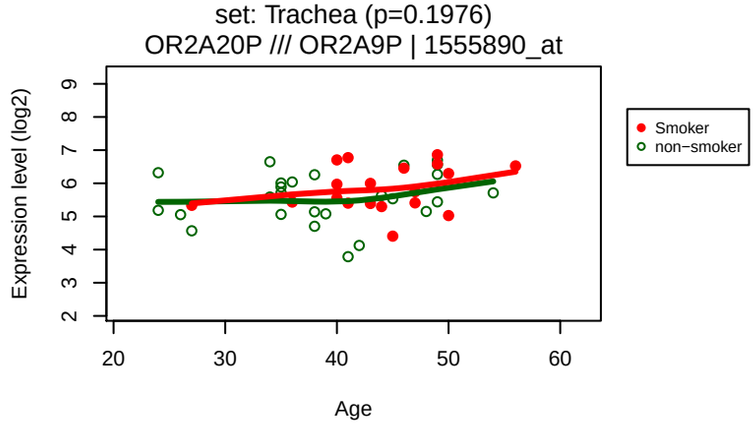


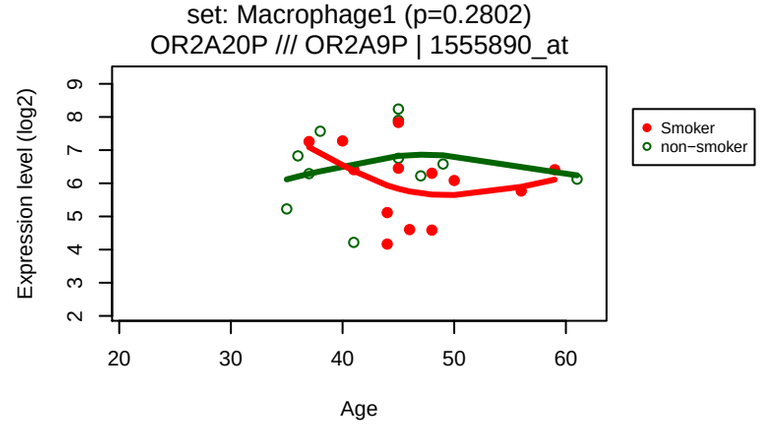

set: SmallAirway1 (p=0.3483)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker

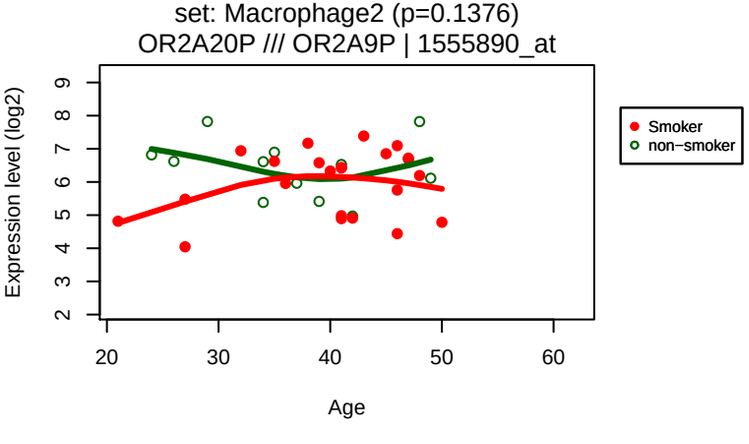

set: SmallAirway2 (p=0.071)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker

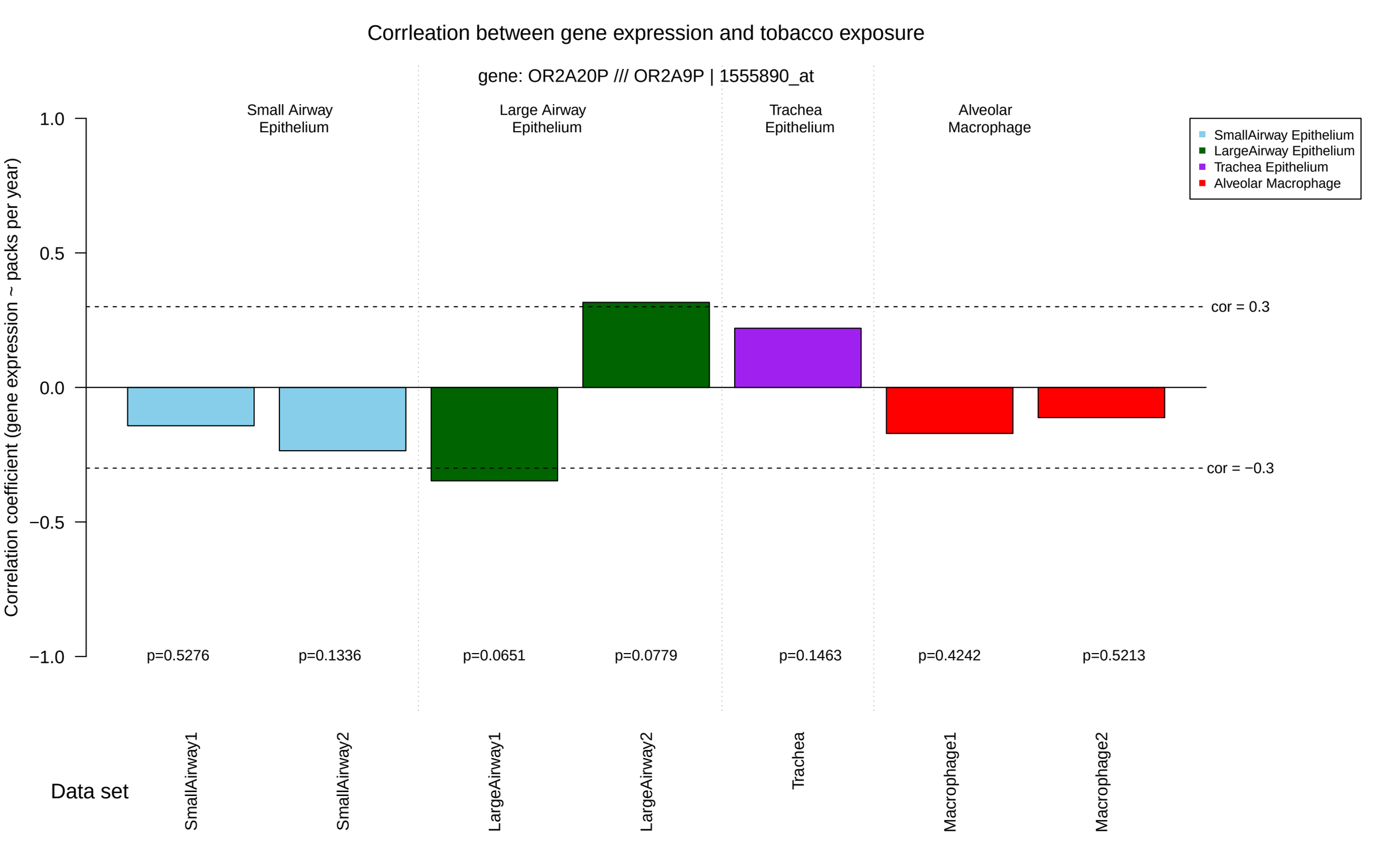

set: LargeAirway1 (p=0.0447)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.381)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1976)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2802)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1376)
OR2A20P /// OR2A9P | 1555890_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2A20P /// OR2A9P | 1555890_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5276
p=0.1336
p=0.0651
p=0.0779
p=0.1463
p=0.4242
p=0.5213
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2